

Procedural Lab Competency Assessment Checklist

Chapter 15: Parenteral Nutrition Preparations

Supplies

- Calculator (basic)
- Pen
- TPN medication order in large, clear-plastic, resealable bag
- TPN label in small, clear-plastic, resealable bag
- D₇₀W, 2000 mL (may substitute any volume bag size ≥ 500 mL)
- Aminosyn 10%, (AA 10%), 500 mL (may substitute any volume bag size ≥ 500 mL)
- SW, 1000 mL; may substitute with any volume ≥ 500-ml bag
- EEC, 1000-mL
- Regular needles (nonfilter, nonvented needles, either 18 or 19 gauge)
- Sterile, 70% IPA swabs, individually packaged × 9
- Potassium phosphate (KPO₄), 3 Mm/mL vial
- Sodium chloride (NaCl), 4 mEq/mL vial
- Potassium chloride (KCl), 2 mEq/mL vial
- Trace minerals (TM), 5-mL vial
- MVI, 10-mL vial
- Humulin R (regular) insulin, 100 units/mL vial
- Calcium gluconate (CaGl), 0.48 mEq/mL vial
- 1-mL Luer-lock syringe
- 5-mL Luer-lock syringes × 2
- 10-mL Luer-lock syringes × 3
- 30-mL Luer-lock syringe
- Hemostats × 3
- IV tubing × 3
- Tamper-evident, sterile, adhesive seals
- Aseptic cleaning wipes presaturated with an appropriate cleaning solution
- Transport vehicle (optional)
- Aseptic garb, including shoe covers, hair cover, face mask, and beard cover (if appropriate)
- Sterile gown
- Sterile, foamed 70% IPA
- Sterile, powder-free gloves
- Scissors
- Paper towels
- Access to sink appropriate for base component fluid disposal
- Horizontal laminar airflow hood
- Permanent, felt-tip marker
- Waste container
- Sharps container

Basic Instructions

- Perform any necessary calculations and then gather the supplies from the supply list.
- Once your instructor gives you permission to begin, garb in the proper order to the point that you would prior to washing your hands.
- Cleanse your hands and forearms with sterile, foamed 70% IPA and then don a sterile gown.
- Finally, don sterile gloves.

Note: For the purposes of this process validation, you will not perform hand-washing or hood-cleaning procedures.

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	Yes	No	NA
TPN Label Verification			
Compared TPN order to TPN label to verify accuracy of all information	☐	☐	☐
Dosage Calculation and Supply Determination			
Reviewed the TPN label to determine the desired concentration and volumes	☐	☐	☐
Correctly performed the mathematical dosage calculations	☐	☐	☐
Gathered the correct supply items	☐	☐	☐
Anteroom Preparations			
Wiped down transport vehicle and all supply items properly	☐	☐	☐
Donned aseptic shoe covers, hair cover, face mask, and beard cover (if appropriate) in the proper order	☐	☐	☐
Donned sterile gown without contaminating it	☐	☐	☐
Clean Room Procedures and TPN Preparation			
Placed supplies in the proper area in the clean room	☐	☐	☐
Used sterile, foamed 70% IPA prior to donning sterile gloves	☐	☐	☐
Opened all alcohol swabs within the DCA	☐	☐	☐
Swabbed the rubber tops of EEC and vials with sterile 70% IPA swab	☐	☐	☐
Prepared the correct volume of each base component and had pharmacist verify base components prior to proceeding	☐	☐	☐
Maintained uninterrupted airflow to the critical sites of all supply items throughout injection, withdrawal, and needle-removal procedures	☐	☐	☐
Did not touch the critical sites of the needle or syringe at any time	☐	☐	☐
Did not touch critical site of each vial without reswabbing site prior to needle insertion	☐	☐	☐
Did not place any syringe onto the hood surface without first attaching a capped needle	☐	☐	☐
Used proper needle insertion technique into the vials	☐	☐	☐
Withdrew the proper volume from each vial	☐	☐	☐
Used proper technique to remove bubbles from syringes	☐	☐	☐
Placed syringes and vials on the hood properly in preparation for verification check by pharmacist or instructor	☐	☐	☐
Used correct solutions and repositioned items in the hood so that the syringes, tubing spike, and EEC received uninterrupted airflow from the HEPA filter at all times	☐	☐	☐
Used proper technique to insert tubing spike into bags	☐	☐	☐
Used proper technique when removing the needle-and-syringe unit from vials	☐	☐	☐
Properly affixed the TPN label to the EEC	☐	☐	☐
Applied a tamper-evident, sterile, adhesive seal over the EEC vial top	☐	☐	☐
Disposed of sharps and nonsharps waste in proper receptacles	☐	☐	☐
Examined EEC for solution clarity at appropriate times	☐	☐	☐
Completion Activities Performed Upon Return to the Anteroom			
Initialed the TPN label in the appropriate place	☐	☐	☐
Returned any remaining supplies to their proper locations	☐	☐	☐
Destroyed base components according to instructions	☐	☐	☐
Removed PPE according to standard procedures	☐	☐	☐

Instructor Feedback

Date

Student/Trainee

Trainer Name

License or Registration Number

Trainer Signature

State Licensed